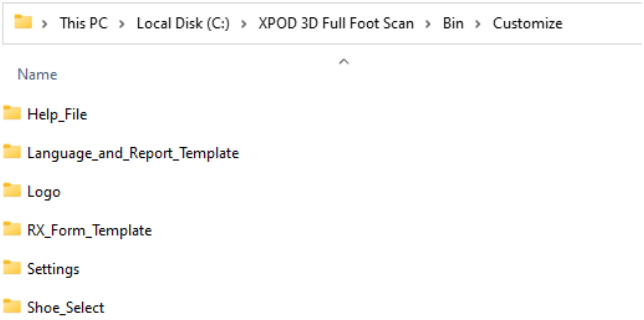
The User Guide is the Help inside the software. Application Guide covers customization and callout integration and is available to developers. Read the User Guide and FactSheet first.

1. Software Customization

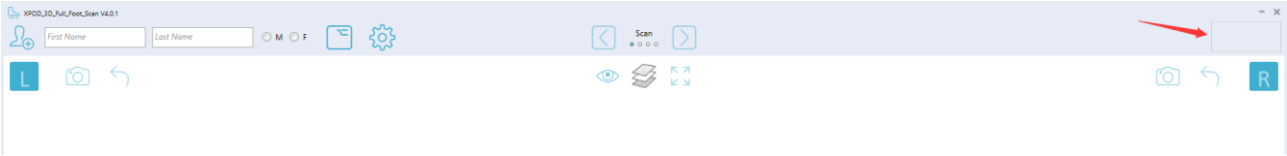
After XPOD software is installed, there are 6 folders under Bin\Customize folder.



Help_File: Same as the User Guide. You can replace with your own PDF help document.

Language_and_Report_Template:
Read "Instructions_Edit_Report_Template.pdf" under
C:\XPOD 3D Full Foot Scan\Bin\Customize\Language_and_Report_Template
to learn edit/translate/customize the interface language and report templates.

Logo: Appears on upper-right corner of scan UI interface. Replace it with your logo. PNG file with exact 165X74 (Horizontal X Vertical) pixel count. The default logo is blank.



Note the logo in the report templates are static images (you can replace at will) and not related to the logo in this folder. Refer to the previous section "Language and report template."

RX_Form_Template: A default RX-Form in simple HTML is included. You can replace with your custom form. The Auto-Filled by software section you can copy and include, or skip. We use Microsoft ActiveX WebBrowser control, so HTML form will be displayed slightly different from a standard browser.

Settings: Changes made in "Settings" of scan and convert are saved in XML file. You can save this preferred XML file. Then after each new install, copy-in this preferred XML, and all "Settings" item will be exactly what you like.

Edit the XML file under "Settings" to change "webpage" to your own. Click on the logo in the scan interface and your website comes up.

If the option "Open Website after Save" is checked in Settings, fill in your own website to be opened after scan file is saved. Display your own web format foot report. Or display insole/shoe products recommended for this customer's foot measurement and analysis. Linking to external website provides unlimited possibilities.

Shoe_Select: Contains definition files for shoe-foot match and jpg file for shoes. Definition file defines length/girth of the foot relative to shoes for good match. Currently only Chinese, EU

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(French) and US shoe size system are supported. We can add UK/JP standards soon.
We can add your own custom shoe sizing system. To include your own shoe match definition file, we need to receive from you the fitting rules of girth/length of foot to your shoe size system. Displayed name of each shoe style is the same as the definition file name; and the jpg file must have the exact same name. You can change the jpg file yourself. Shoe_Select is supported by Command-Line callout, but not EXE or Rocket callout.

After each software upgrade, the language/settings items may have changed/increased. Remember to change your custom files accordingly. If you copy in custom language or settings file for older version software, new version software may crash.

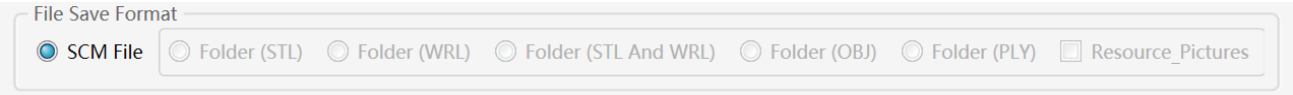
Each time you install standard software on a computer, replace with your customized contents to have your own custom software. If you find this troublesome, we can provide a custom software installation exe with your custom data embedded.

Foreign Language Restrictions: XPOD software installation path may contain only English with no foreign language characters. On a computer with foreign language A version Windows, language A is fully supported. If you later install a language pack of foreign language B, then language B may not be fully supported. English is always supported.

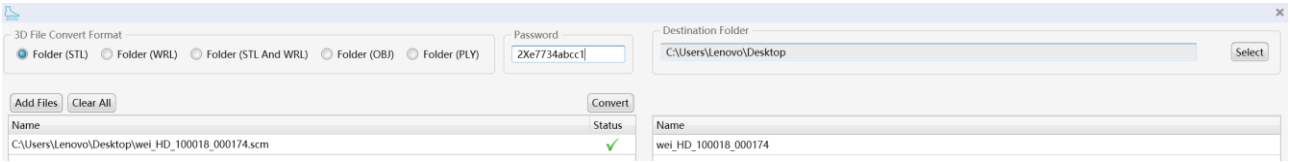
2. Scanner Encryption Lock

If our direct customers deploy scanners below cost, they should have control over STL/WRL 3D files for CAD design. Scanner encryption achieves that purpose.

Encrypted scanners can only save as SCM file (cannot save as STL/WRL/OBJ/PLY). Under "Settings" folder options (STL/WRL/OBJ/PLY) are not available.



"Convert" section will demand a password before conversion can proceed. Encryption code is 12 digits, can be any letter, number, or symbol combinations. Non-encrypted scanners have default password "000000000000".



Once the correct password is entered, conversion may proceed. Click on converted file to view folder with converted files. Scanners can be encrypted ONLY before shipping.

3. Command-Line Callout

Our customers often have proprietary CAD design or data management system and wish to obtain data directly. Command-line is the easiest route to that end.
Before a command-line session, close the XPOD main software.
Command-line draws parameters from settings, which is saved to the XML file under C:\XPOD 3D Full Foot Scan\Bin\Customize\Settings. XML file also contains the "File Convert Format" and

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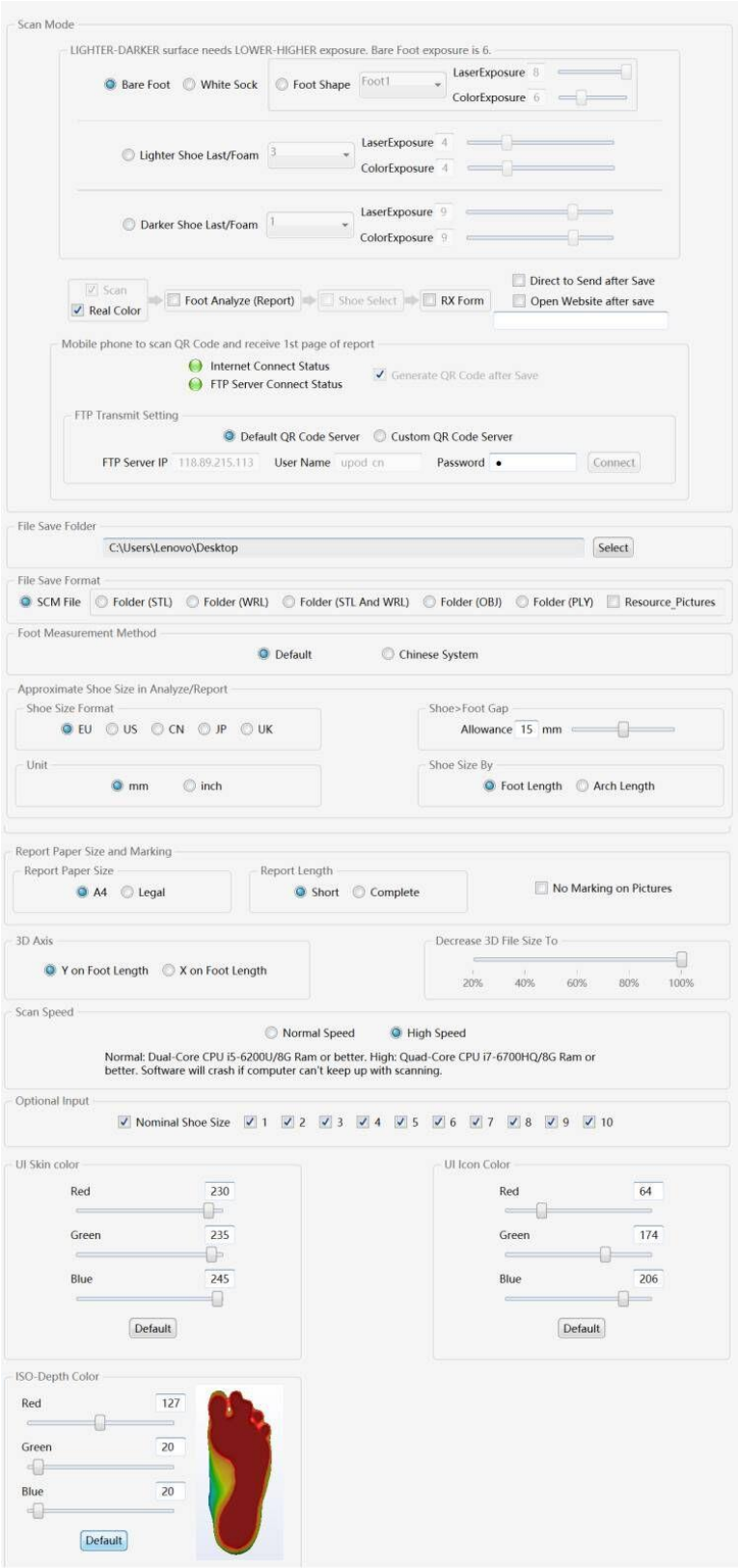
"Convert Destination Folder" of the "Convert" function. By changing the XML file, you can control settings without opening the XPOD software.




```
▼<Body>
  ▼<SettingInformation>
    ▼<SettingData>
      <WebPage> </WebPage>
      <SkinRed>230</SkinRed>
      <SkinGreen>235</SkinGreen>
      <SkinBlue>245</SkinBlue>
      <IconRed>64</IconRed>
      <IconGreen>174</IconGreen>
      <IconBlue>206</IconBlue>
      <ISODepthRed>127</ISODepthRed>
      <ISODepthGreen>20</ISODepthGreen>
      <ISODepthBlue>20</ISODepthBlue>
      <FileSaveFolder>C:\Users\Lenovo\Desktop</FileSaveFolder>
      <FileSaveFormat>0</FileSaveFormat>
      <FootMeasurementMethod>0</FootMeasurementMethod>
      <ResourcePictures>0</ResourcePictures>
      <ScanMode>0</ScanMode>
      <ReportPaperSize>0</ReportPaperSize>
      <ReportLength>0</ReportLength>
      <NoMarkingOnPicture>0</NoMarkingOnPicture>
      <!-- Bare Foot ScanMode = 0 -->
      <!-- White Sock ScanMode = 1 -->
      ▼<FootShapeExposureSetting>
        <!-- FootShape ScanMode = 2 -->
        <ActiveFootShapeIndex>0</ActiveFootShapeIndex>
        <LaserExposure1>8</LaserExposure1>
        <!-- Range 5 to 8 -->
        <ColorExposure1>6</ColorExposure1>
        <!-- Range 5 to 8 -->
        <LaserExposure2>5</LaserExposure2>
        <!-- Range 5 to 8 -->
        <ColorExposure2>5</ColorExposure2>
        <!-- Range 5 to 8 -->
        <LaserExposure3>5</LaserExposure3>
        <!-- Range 5 to 8 -->
        <ColorExposure3>5</ColorExposure3>
        <!-- Range 5 to 8 -->
      </FootShapeExposureSetting>
      ▼<LighterObjectExposureSetting>
        <!-- Lighter Object ScanMode = 3 -->
        <ActiveLighterObjectIndex>2</ActiveLighterObjectIndex>
        <LaserExposure1>5</LaserExposure1>
        <!-- Range 1 to 12 -->
        <ColorExposure1>5</ColorExposure1>
        <!-- Range 1 to 12 -->
        <LaserExposure2>4</LaserExposure2>
        <!-- Range 1 to 12 -->
        <ColorExposure2>4</ColorExposure2>
        <!-- Range 1 to 12 -->
        <LaserExposure3>4</LaserExposure3>
        <!-- Range 1 to 12 -->
        <ColorExposure3>4</ColorExposure3>
        <!-- Range 1 to 12 -->
        <LaserExposure4>4</LaserExposure4>
        <!-- Range 1 to 12 -->
        <ColorExposure4>4</ColorExposure4>
        <!-- Range 1 to 12 -->
        <LaserExposure5>4</LaserExposure5>
        <!-- Range 1 to 12 -->
        <ColorExposure5>4</ColorExposure5>
        <!-- Range 1 to 12 -->
      </LighterObjectExposureSetting>
      ▼<DarkerObjectExposureSetting>
        <!-- Darker Object ScanMode = 4 -->
        <ActiveDarkerObjectIndex>0</ActiveDarkerObjectIndex>
        <LaserExposure1>9</LaserExposure1>
        <!-- Range 1 to 12 -->
        <ColorExposure1>9</ColorExposure1>
        <!-- Range 1 to 12 -->
        <LaserExposure2>9</LaserExposure2>
        <!-- Range 1 to 12 -->
        <ColorExposure2>9</ColorExposure2>
        <!-- Range 1 to 12 -->
        <LaserExposure3>9</LaserExposure3>
        <!-- Range 1 to 12 -->
        <ColorExposure3>9</ColorExposure3>
        <!-- Range 1 to 12 -->
        <LaserExposure4>9</LaserExposure4>
        <!-- Range 1 to 12 -->
        <ColorExposure4>9</ColorExposure4>
        <!-- Range 1 to 12 -->
        <LaserExposure5>5</LaserExposure5>
        <!-- Range 1 to 12 -->
        <ColorExposure5>9</ColorExposure5>
        <!-- Range 1 to 12 -->
      </DarkerObjectExposureSetting>
      <FootAnalyze>0</FootAnalyze>
      <ShoeSelect>0</ShoeSelect>
      <QRCode>0</QRCode>
      <QRCodeFtpServerType>0</QRCodeFtpServerType>
      <RXForm>0</RXForm>
      <RealColorTexture>1</RealColorTexture>
      <DirectToSendAfterSave>0</DirectToSendAfterSave>
      <OpenWebsiteAfterSave>0</OpenWebsiteAfterSave>
      <Axis3D>0</Axis3D>
      <Decrease3DFileSizeTo>100</Decrease3DFileSizeTo>
      <ShoeSizeFormat>0</ShoeSizeFormat>
      <ShoeFootGap>15</ShoeFootGap>
      <Unit>0</Unit>
      <ShoeSizeBy>50</ShoeSizeBy>
      <File3DConvertFormat>0</File3DConvertFormat>
      <ConvertDestinationFolder>C:\Users\Lenovo\Desktop</ConvertDestinationFolder>
      <HighSpeed>1</HighSpeed>
      <HighPrecision>1</HighPrecision>
      <ExternalLightFiltering>0</ExternalLightFiltering>
      <NominalShoeSize>1</NominalShoeSize>
      <OptionalInput1>1</OptionalInput1>
      <OptionalInput2>1</OptionalInput2>
      <OptionalInput3>1</OptionalInput3>
      <OptionalInput4>1</OptionalInput4>
      <OptionalInput5>1</OptionalInput5>
      <OptionalInput6>1</OptionalInput6>
      <OptionalInput7>1</OptionalInput7>
      <OptionalInput8>1</OptionalInput8>
      <OptionalInput9>1</OptionalInput9>
      <OptionalInput10>1</OptionalInput10>
    </SettingData>
  </SettingInformation>
</Body>
```


Most of the items are 0/1 and self-evident. Only the following need clarification:

<FileSaveFormat> "0" is SCM. "1" is Folder (STL). "2" is Folder (WRL). "3" is Folder (STL and WRL, "4" is for Folder (OBJ), and "5" is for Folder (PLY))

<ScanMode> "0" is Bare Foot. "1" is White Sock. "2" is Foot Shape. "3" is Lighter Shoe Last/Foam. "4" is Darker Shoe Last/Foam.

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<LaserExposure> and <ColorExposure> are full numbers from 1 to 12.
Foot Shape exposure range 5-8;
Lighter Shoe Last/Foam exposure range 1-12.
Darker Shoe Last/Foam exposure range 1-12
Darker surface requires higher exposure values, and lighter surface requires lower.
We provide two commonly used scan mode "Bare Foot" and "White Sock" with fixed exposure settings. "Foot Shape" allows exposure adjustment, and is for other sock color, very dark skin, or when the fixed exposure doesn't work well. Only foot shape allows analyze/measurement/report. Non-Foot shape cannot automatically locate landmark points, hence analyze is disabled.

"Resource Pictures" option enables output all the pictures from the default PDF report. "No Marking on Pictures" gives pictures without landmark points or measurement lines.

Feedback values are saved to the clipboard. If you attempt to read the clipboard when our program is writing to it, you may get an error message. Your program needs to account for this and read only after writing is completed.

Command-line callout "Scan"
The complete callout lines (2 lines) would be:
1 cd C:\XPOD 3D Full Foot Scan\Bin
2 "C:\XPOD 3D Full Foot Scan\Bin\XPOD_3D_Full_Foot_Scan.exe" -scan John Doe 0 000000000000
-scan is for Scan
John Doe is First Name and Last Name
0 is Male; change to 1 for Female
000000000000 means this scanner is not encryption locked. If this scanner is encrypted, and correct password is provided here, output file format will be what is selected in settings; if password is not correct, output file will be SCM regardless of settings.

The above callout saves John_Doe_030021_000025 to location set in the XML file, where "030021"is Scanner Serial No. and "000025" is Scan Count (number of saves). Since it may take multiple scans before save, the final file names are unknown at callout.
To resolve this problem, we added an option at the end of the scan callout line:
-name d:\\any folder\\any name you want
When this option is added, you can name the output file/ folder anything, and save to any folder location. The complete callout lines (2 lines) would be:
1 cd C:\XPOD 3D Full Foot Scan\Bin
2 "C:\XPOD 3D Full Foot Scan\Bin\XPOD_3D_Full_Foot_Scan.exe" -scan John Doe 0 000000000000 -name d:\\any folder\\any name you want

To support mobile apps, 3D file size needs to be much smaller.
-mobile
When this option is added, output an additional small 3D file (sparse mesh) in STL/WRL/OBJ/PLY. Note that only when STL/WRL/OBJ/PLY is selected in settings, this option is active. If SCM file format is selected, this option is disabled. And WRL/OBJ/PLY sparse mesh will not contain texture.
The complete callout lines (2 lines) would be:
1 cd C:\XPOD 3D Full Foot Scan\Bin

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2 "C:\XPOD 3D Full Foot Scan\Bin\XPOD_3D_Full_Foot_Scan.exe" -scan John Doe 0 000000000000 -mobile

The following value is saved to clipboard for the corresponding status:
0 wrong format of the callout line
1 Scanner not found
3 Scan interface comes up, ready to scan.
4 Program exited without saving
5 Program exited with saving
7 Program existed without scan

The scan interface comes up, "First Name", "Last Name", "M/F Gender" and "Settings" are all locked and cannot be changed. Click "Scan" button or press foot switch to scan. Click "Save" to save to "File Save Folder" specified in the XML file.

After you click "Save", software interface will close down, as a scan session is completed. The exception is if you selected "Generate QR code after save", software interface will wait for you to click the upper-right corner "X" to close down. This allows the user to wait for report (1st page) to upload, and scan QR code with phone to obtain 1st page of report.

Make sure you can write to the save folder. XPOD software may crash or non-perform if you attempt to write to a location without save privilege (Windows User Access Control).

Command-line callout "Scanner Serial No. and Scan Count"
The complete callout lines (2 lines) would be:
1 cd C:\XPOD 3D Full Foot Scan\Bin
2 "C:\XPOD 3D Full Foot Scan\Bin\XPOD_3D_Full_Foot_Scan.exe" –machine
-machine is for obtaining Scanner Serial No. and Scan Count (number of saves)
Obtain Scanner Serial No.
XPOD-S has serial no. "1XXXXX", and XPOD-SS has serial no. "2XXXXX".
XPOD-SS2 has serial no. "3XXXXX"

The following value is saved to clipboard for the corresponding status:
1 scanner not found

Command-line callout "Open"
The complete callout lines (2 lines) would be:
1 cd C:\XPOD 3D Full Foot Scan\Bin
2 "C:\XPOD 3D Full Foot Scan\Bin\XPOD_3D_Full_Foot_Scan.exe" –open C:\John_Doe_030021_000021.scm C:\ John_Doe_030021_000021New.scm
-open is for open
C:\John_Doe_030021_000021.scm is the SCM file to be opened
C:\ John_Doe_030021_000021New.scm is the SCM file name to be saved
The following value is saved to clipboard for the corresponding status:
0 wrong format of the callout line

Open a SCM file, make changes, click "Save" to exit or just close the interface.

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Command-line callout "Convert"

The complete callout lines (2 lines) would be:

1 cd C:\XPOD 3D Full Foot Scan\Bin

2 "C:\XPOD 3D Full Foot Scan\Bin\XPOD_3D_Full_Foot_Scan.exe" –convert C:\John_Doe_030021_000021.scm 000000000000

-convert is for convert

C:\John_Doe_030021_000021.scm is the SCM file to be converted

000000000000 means this SCM file is not encryption locked. If the file is locked, correct password must be provided before convert can proceed.

Note to obtain measurement and resource pictures at convert, "Analyze" and "Resource Pictures" must be checked in settings during scan.

To support mobile apps, 3D file size needs to be much smaller.

-mobile

When this option is added, you can output an additional small 3D file (sparse mesh) in STL/WRL/OBJ/PLY. And WRL/OBJ/PLY sparse mesh will not contain texture.

The complete callout lines (2 lines) would be:

1 cd C:\XPOD 3D Full Foot Scan\Bin

2 "C:\XPOD 3D Full Foot Scan\Bin\XPOD_3D_Full_Foot_Scan.exe" –convert C:\John_Doe_030021_000021.scm 000000000000 -mobile

The following value is saved to clipboard for the corresponding status:

0 wrong format of the callout line

2 password incorrect

5 convert successful

Convert one file per callout. <3D File Convert Format> and <Destination Folder> are set in the XML settings file.

4. EXE Callout

The command-line callout still brings up our software interface. To avoid our software interface, EXE callout can access our scanner, take scans, and return the files.

The EXE callout supports the most features but requires intimate knowledge of the scan workflow. You'll probably need our help to make it work. We highly recommend the Rocket Callout (next section) which is much simpler to implement.

Before an EXE callout session, close the XPOD software.

Exe callout draws parameters from settings XML file under C:\XPOD 3D Full Foot Scan\Bin\Customize\Settings. Refer to Command-Line section. You can edit the settings XML file **ONLY before** starting XPOD Proxy. Don't edit the settings XML when the proxy is active. You also need to ensure the file save folder exist and you can write to it. Otherwise, our software will write the files to the desktop.

Demo codes in C#, C++, and NodeJS is located at:

C:\XPOD 3D Full Foot Scan\CallProxy_DEMO

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Start XPOD Proxy (System tray will show the XPOD Proxy icon)
Run C:\XPOD 3D Full Foot Scan\Bin\XPOD_Proxy.exe, with Start in directory "C:\XPOD 3D Full Foot Scan\Bin".

The complete callout lines (2 lines) would be:
1 cd C:\XPOD 3D Full Foot Scan\Bin
2 "C:\XPOD 3D Full Foot Scan\Bin\XPOD_Proxy.exe" -Lwd XXX -Rwd YYY -Lwd2 AAA -Rwd2 BBB
-Lwd XXX Bounding Left 3D view window (XXX is the HWND)
-Rwd YYY Bounding Right 3D view window (YYY is the HWND)
-Lwd2 AAA Bounding Left 3D Analyze window (AAA is the HWND)
-Rwd2 BBB Bounding Right 3D Analyze window (BBB is the HWND)
This allows you to display 3D with our 3D view window.

Command-line Parameters for each operation
The complete callout lines (2 lines) would be:
1 cd C:\XPOD 3D Full Foot Scan\Bin
2 "C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" –enable
1)-enable
"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" –enable
Enable Scanner
In the standard XPOD software, you must enter name and gender to activate the scan button. With -enable here you have the same effect. Foot switch is also activated.
101 Start
201 End

2)-scan
"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" -scan -l John Doe 0 000000000000
Scan
-l (Left Foot)
-r (Right Foot)
Note this is for XPOD-S. For XPOD-SS this becomes -l (Left and Right Foot)
John Doe is First Name and Last Name
0 is Male; change to 1 for Female
000000000000 means this scanner is not encryption locked. If this scanner is encrypted, and correct password is provided here, output file format will be what is selected in settings; if password is not correct, output file will be SCM regardless of settings.
102 Start
202 End
401 3D Reconstruction failed and end. This is equivalent to having an empty 3D window in XPOD software. This is caused by no object in the scan area, object being reflective or exposure setting not correct for object color, or hardware problem.
402 Frame drop and end. This is equivalent to Code 1 error in XPOD software. This is caused by PC not able to keep up with scan, or hardware problems.

-stream_stl
"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" -scan -l John Doe 0 000000000000 -stream_stl

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After 3D reconstruction, save STL file (normal and sparse-mesh) to Bin\Proxy_STL folder
-l (Left Foot): Bin\Proxy_STL\Left.stl; Left_Decimation.stl
-r (Right Foot): Bin\Proxy_STL\Right.stl; Right_Decimation.stl
Note this is for XPOD-S. For XPOD-SS this becomes -l (Left and Right Foot)
Sparse mesh is obtained after decimation. It is a small 3D file for mobile use.
4 status values, same as -scan section
403 save STL file failed
Wait for status value 202 before reading file, or errors can occur.

-name
"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" -scan -l John Doe 0 000000000000 -name d:\\any folder\\any name you want
Name
When this option is added, you can name the file and save folder to anything. This option only set file name and folder for this scan and will NOT change the save folder inside settings XML file.
4 status values, same as -scan section

3) -back
"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" -back -l
Back
-l (Left Foot)
-r (Right Foot)
Note this is for XPOD-S. For XPOD-SS this becomes -l (Left and Right Foot)
If you are not satisfied with this scan, return the scanner to the state after -enable.
103 Start
203 End

4) -newscan
"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" -newscan
Newscan
Scanner back to the state before -enable. Ready for the next scan.
104 Start
204 End

5)-analyze
"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" -analyze
Analyze
Output measurements, landmark point coordinates, and resource pictures (if this is selected in the XML setting file). Analyze only works on foot shape and may crash if attempts are made on other object shapes including foam impression and shoe last.
105 Start
205 End
If you encounter crash during analyze, send back the file under C:\XPOD 3D Full Foot Scan\Bin\Debug_For_Proxy\Last.scm for analysis.

6)-save
"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" -save

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Save

Save scan and analyze result.

106 Start

206 End

-res

"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" -save -res

Res

When saving SCM, also save Resource_Pictures folder and Mesurement.csv to Bin\XPOD_TEMP.

106 Start

206 End

Note Resource_Pictures and Analyze must be checked under settings.

7)-mobile

"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" -save -mobile

Mobile

When this option is added, output an additional small 3D file (sparse mesh) in STL/WRL/OBJ/PLY. Note that only when STL/WRL/OBJ/PLY is selected in settings, this option is active. If SCM file format is selected, this option is disabled. And WRL/OBJ/PLY sparse mesh will not contain texture.

2 status values, same as above section

8)–machine

Machine

"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" –machine

Obtain Scanner Serial No. XPOD-SS has serial no. "2XXXXX".

108 Start

XXXXXX Return serial number and end

-1 Failed to obtain serial number and end. Usually this is hardware problem.

9)–Count

"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" –count

Count

Obtain Scan Count of this XPOD, number of saves (not scans) in 5 digits.

109 Start

XXXXX Return scan count and end

-1 Failed to obtain scan count and end. Usually this is hardware problem.

10)–Quit

"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" –quit

Quit

Close down

107 Start

207 End

11)-Check

"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" –check

Check

Obtain current XPOD_Proxy status.

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There are two ways to obtain XPOD_Proxy status value:

Use 11)-Check Above; or

Read a file in memory "XPOD_Proxy", a 4-byte integer gives the status of the proxy.

EXE Callout			
Error Value	**Category**	**Causes**	**Solutions**
-1	Machine	Failed to obtain serial number and end. Scanner not connected to PC, communication interrupted.	Properly connect the scanner and PC.
-1	Count	Failed to obtain scan count and end. Scanner not connected to PC, communication interrupted.	Properly connect the scanner and PC.
401	Scan	3D Reconstruction failed and end: 1. No object in scan area. 2. The object is reflective. 3. Exposure setting is not correct for object color. 4. Hardware problem.	1. Put an object in scan area. 2. Prepare the powder coating for shiny surface. 3. Reset the exposure values. 4. Check the hardware.
402 (same as Code 1 in XPOD software)		Frame drop and end: 1. CPU cannot keep up with scanning. 2. Use another USB hub to combine two USB heads of USB cable. 3. Other applications requiring high CPU power tie up CPU/RAM. 4. Hardware problems.	1. Minimum CPU i5-1240P, 16G. 2. Directly plug two USB heads of USB cable into PC. 3. Close the running Windows updates or other applications, e.g., ongoing Windows update. 4. Check the hardware.
403		Save STL file failed. Scan not finished.	Wait for status value 202 before reading file.
608	Client	Mutex Process. Start more than two XPOD_Client.exe programs simultaneously.	Start only one XPOD_Client.exe program.
600		Wrong format of the callout line.	Ensure correct format of callout line.
601		Failed to initialize sockets.	Check the network connection and security settings in Windows Security Center.
602		Create socket failed.	
603		Connect proxy failed.	
508	Proxy	Mutex Process. Start more than two XPOD_Proxy.exe programs simultaneously.	Start only one XPOD_Proxy.exe program.
500		License not found (software dongle missing). Encryption dongle not inserted.	Insert the encryption dongle.
501		Failed to initialize sockets.	Check the network connection and security settings in Windows Security Center.
502		Create socket failed.	
503		Socket bind failed.	
504		Socket listen failed.	
505		Camera initialization failed.	Check the connection of power cable and USB cables.
506		CreateFileMapping failed. Memory mapping failed.	Check the memory.
507		MapViewOfFile failed. Memory mapping failed.	

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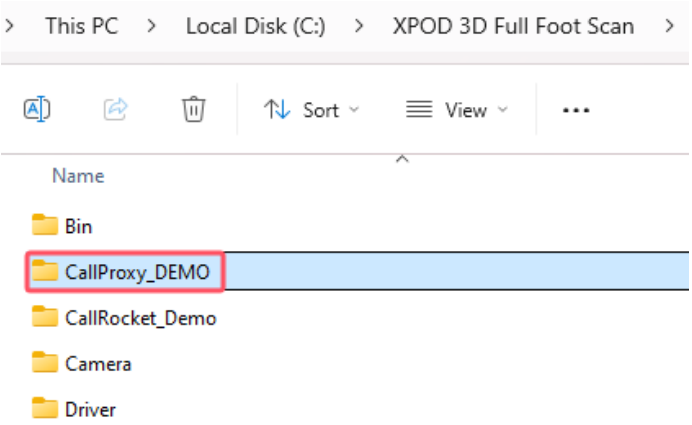
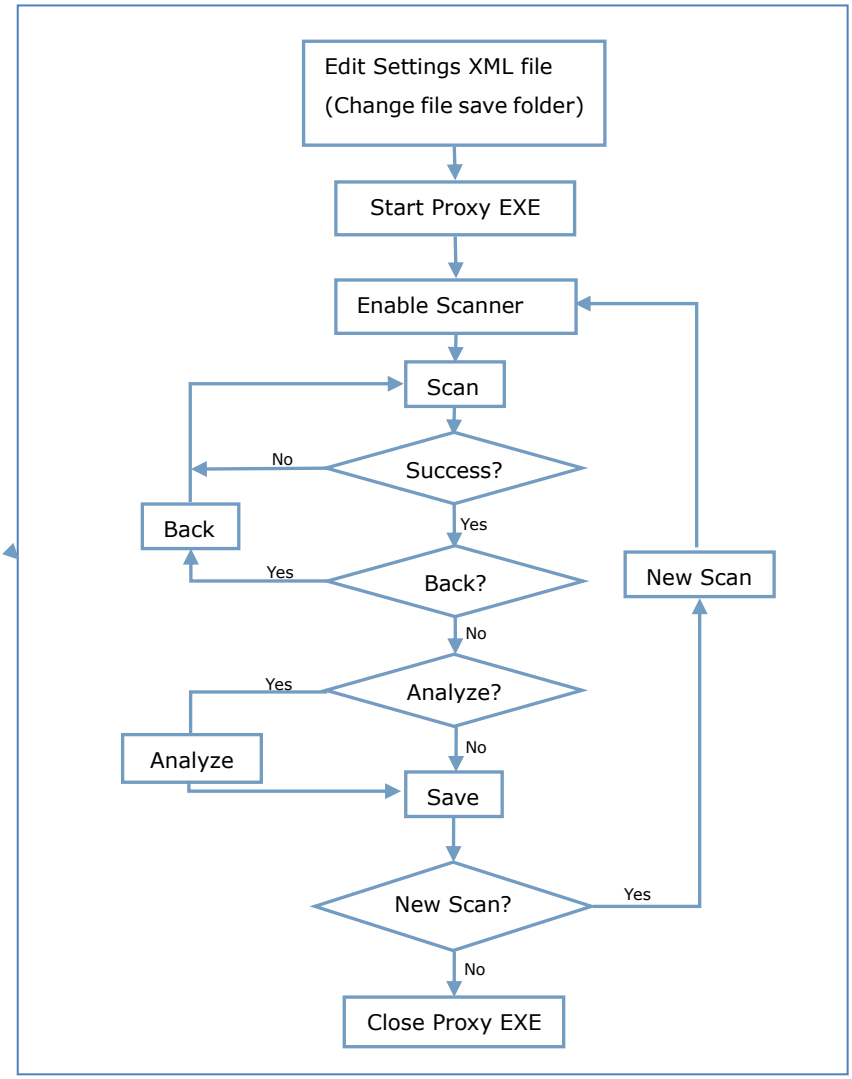
Demo codes in C#, C++, and NodeJS is located at:
C:\XPOD 3D Full Foot Scan\CallProxy_DEMO



Due to the complexity of this callout method. We no longer recommend it. The Rocket callout is much easier to implement. See the next section.

You can edit-save the settings XML file **ONLY before** starting XPOD Proxy. Don't edit the settings XML (to change file save folder) when the proxy is active. You also need to ensure the file save folder exist and you can write to it. Otherwise, our software will write the files to the **desktop**.



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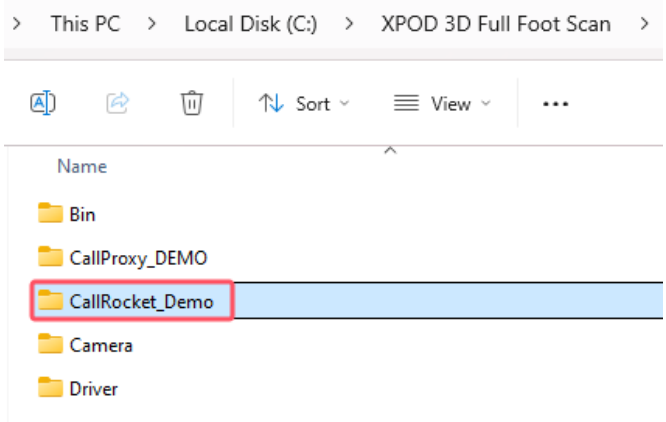
5. Rocket Callout

The command-line callout still brings up our software interface. To avoid our software interface, Rocket callout can access our scanner, take scans, and return the files.

The EXE callout supports the most features but requires intimate knowledge of the scan workflow. You'll probably need our help to make it work. We highly recommend the Rocket Callout (this section) which is much simpler to implement.

Before a Rocket callout session, close the XPOD software.
Rocket callout draws parameters from settings XML file under C:\XPOD 3D Full Foot Scan\Bin\Customize\Settings. Refer to Command-Line section.
Since XPOD V5.2, you can edit the settings XML file before each scan. With previous versions of XPOD software, you can only edit the settings XML file before Starting the scanner.
You also need to ensure the file save folder exist and you can write to it. Otherwise, our software will write the files to the desktop.

Demo codes in C#, C++, Python and NodeJS is located at:
C:\XPOD 3D Full Foot Scan\CallRocket_Demo



Callout Parameters for each operation

1 -start
"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Rocket.exe" -start
Start Scanner

2 -scan
"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Rocket.exe" -scan -l John Doe 0 000000000000
Scan

-l (Left Foot)
-r (Right Foot)
Note this is for XPOD-S. For XPOD-SS this becomes -l (Left and Right Foot)
John Doe is First Name and Last Name
0 is Male; change to 1 for Female
000000000000 means this scanner is not encryption locked. If this scanner is encrypted, and correct password is provided here, output file format will be what is selected in settings; if password is not correct, output file will be SCM regardless of settings.
-name
"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Client.exe" -scan -l John Doe 0 000000000000 -name

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d:\\any folder\\any name you want

Name

When this option is added, you can name the file and save folder to anything. This option only set file name and folder for this scan and will NOT change the save folder inside settings XML file.

3 -save

"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Rocket.exe" -save

Save scan and analyze result.

4 -exit

"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Rocket.exe" -exit

Close down

5 -machine

"C:\XPOD 3D Full Foot Scan\Bin\XPOD_Rocket.exe " -machine

Get the serial number of the scanner

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Rocket Callout		
Status Value	**Description**	**Remarks**
0	System fatal error	Windows system error.
1	Run Rocket without arguments	Check the command line parameters.
2	Run Rocket with invalid arguments	Check the format of command line parameters.
3	Rocket is busy	Rocket is processing previous command, it can accept next command after processing.
4	Another Rocket is running	No need to restart Rocket.
5	No Rocket is running	Start Rocket and then send the commands.
6	No found control board and cameras	Check the connection of power cable and USB cables.
7	No found control board	Check the connection of power cable and USB cables.
8	No found cameras	Check the connection of power cable and USB cables.
9	Start Rocket successful	Can send the command.
10	Request Rocket to scan foot	Wait for scan to complete.
11	Scan foot failed	Check to find causes.
12 (same as Code 1 in XPOD software)	Scan foot failed due to camera frame dropping	1. Minimum CPU i5-1240P, 16G. 2. Directly plug two USB heads of USB cable into PC. 3. Close the running Windows updates or other applications, e.g., ongoing Windows update. 4. Check the hardware.
13	Scan foot successful	
14	Request Rocket to save file	Wait for operation to complete.
15	Save failed because the save path does not exist	Check whether save path exists.
16	Save failed because scan data does not exist	1. Put an object in scan area. 2. Prepare the powder coating for shiny surface. 3. Reset the exposure values. 4. Check the hardware.
17	Save file successful	
18	Request Rocket to exit	Wait for operation to complete.
19	Rocket exited	
20	Request Rocket to get serial no.	Wait for operation to complete.
21	Request Rocket to scan left/right foot	Wait for operation to complete.
22	Set Rocket to scan left/right foot	Command completed.

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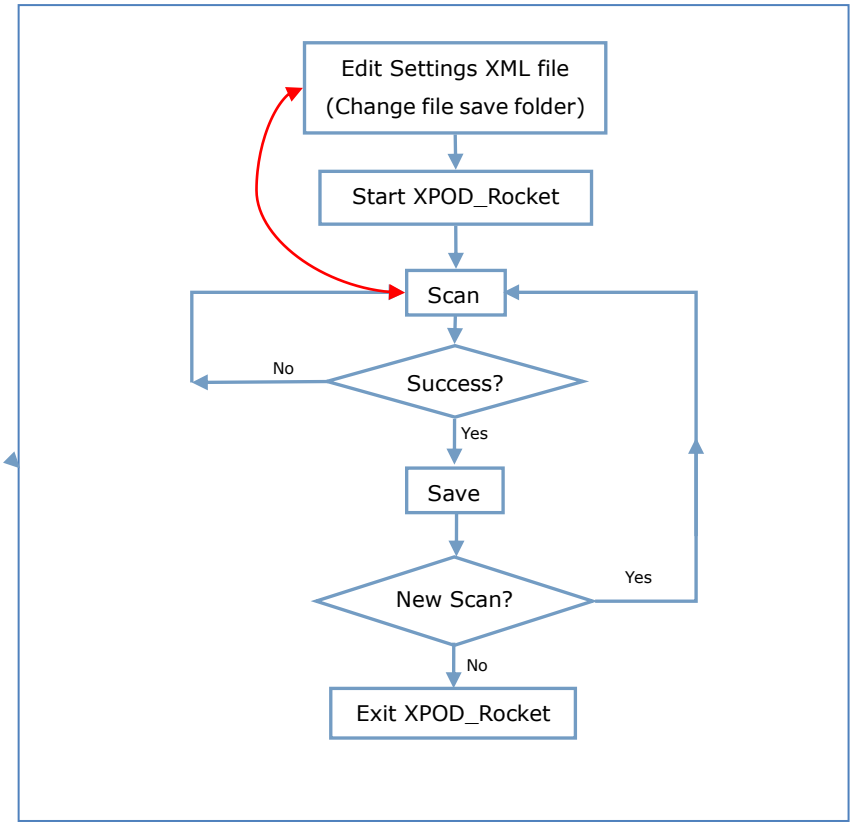

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